

Qindong Gan

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Address: Chuo City, Tokyo, Japan

EDUCATION

Waseda University

Tokyo, Japan

Master of Engineering in Computer Science and Communications Engineering

09/2025-08/2027 (Expected)

- **GPA: 3.33/4.00 (Waseda scale; A+ in 8 of 12 completed courses)**

University of Cincinnati

Cincinnati, OH, US

Bachelor of Science in Electrical Engineering

09/2020-05/2025

- **GPA: 3.571/4.000**

Chongqing University

Chongqing, China

Bachelor of Engineering in Electrical Engineering and Automation

09/2020-06/2025

- **GPA: 3.35/4.00**

RESEARCH OUTPUTS

- Q. Gan, Z. Pan, T. D. Ho, J. Wu, and S. Shimamoto, "Causally Verified QoS-Floor Constraints for Decentralized SAGIN Access Association." Manuscript submitted to IEEE CCNC 2027; under review.
Research problem and finding: Examined whether decentralized Lagrangian control could enforce per-slot QoS floors under interference coupling. In simulation, the revised expected-shortfall formulation raised the floor 17.1x at $N = 12$ and 63.1x at $N = 20$ over matched unconstrained MAPPO.
First-author contribution: Formulated the problem and constraint, built the EPyMARL environment and baselines, designed the intervention protocol, and led the experiments and analysis.
- Q. Gan, "ShinsaEvolve: Independent Audits in LLM-Driven Recipe Evolution." Technical report and public research artifact, 2026.
[GitHub](#) | [Technical Report](#)
Research problem and finding: Examined whether self-reported training scores could guide LLM-driven program evolution. Candidate-shaped rewards amplified selection errors. Fixed raw rewards removed score saturation, while training-to-audit misranking persisted.
Independent contribution: Designed the audit protocol, implemented the system, and ran a 2x2 LunarLander study with 12 outer-loop runs and 180 candidates.

RESEARCH EXPERIENCE

Causally Verified QoS-Floor Constraints for Decentralized SAGIN Access Association

03/2026-Present

First Author | Supervised by Shigeru Shimamoto, Waseda University

- Formulated vehicle-side access association in a three-tier SAGIN as an interference-coupled constrained Markov game and implemented MAPPO under centralized training and decentralized execution.
- Diagnosed a nonpositive aggregate residual and traced an apparent 4.4x floor gain to per-episode multiplier resets while the constraint signal remained inactive.
- Designed cut, cross-agent misdirection, and sign-reversal interventions to test whether learned multipliers carried the intended constraint signal.
- Reformulated the constraint as a one-sided expected-shortfall budget and evaluated performance, outage, re-association, and dwell-time tradeoffs against six baselines.

ShinsaEvolve: Independent Audits in LLM-Driven Recipe Evolution

08/2026-Present

Independent Research Project and Technical Report | [GitHub](#) | [Technical Report](#)

- Developed an audit-gated LLM recipe-evolution system that evaluates trained policies on fresh seeds with harness-controlled raw rewards and keeps candidate recipe code outside the audit path.
- Compared candidate-reported training performance with independent audit performance and measured how reward design changed archive selection across the replicated study.

RGB-Thermal Semantic Segmentation for Autonomous Driving

07/2025-08/2025

Full-time AI Algorithm Intern | Supervised by Jiachen Liang, SERES Group Co., Ltd.

- Adapted DFormerV2 for RGB-thermal inputs and designed multimodal fusion modules for autonomous-driving semantic segmentation.
- Built the training and evaluation workflow and conducted comparative and ablation studies on the project dataset.

Bio-inspired Model of Mapping and Navigation for Robots

09/2024-05/2025

Research Assistant | Supervised by Ali Minai, University of Cincinnati

- Implemented visual-data preprocessing and CNN-based feature extraction for a navigation model using place-cell, boundary-vector-cell, and head-direction-cell representations.
- Conducted multimodal fusion experiments, tuned model parameters, and analyzed representation stability and navigation performance in simulation.

PRESENTATIONS

- Presented "Multi-Agent Reinforcement Learning for Intelligent Access Management in Space-Air-Ground Integrated Networks" at the ASPIRE-CRC-TCA International Workshop, Memorial University, August 2026.

HONORS AND AWARDS

- Best Master's Student Presentation Award, ASPIRE-CRC-TCA International Workshop, Memorial University, 2026.
- Dean's List, University of Cincinnati, Fall 2024 and Spring 2025.
- Cooperative Education & Practice - Excellence, University of Cincinnati.
- Second Prize, National College Student Robot Competition; Outstanding Student Scholarship (Top 10%); Outstanding Graduate, Chongqing University.

TEACHING EXPERIENCE

Teaching Assistant, University of Cincinnati

09/2024-05/2025

- Led recitation and lab support for an undergraduate engineering course in English; graded assignments and examinations and reported recurring comprehension gaps to the instructor.

SKILLS

Programming and Tools: Python, MATLAB, PyTorch, NumPy, pandas, scikit-learn, EPyMARL, MATLAB Communications Toolbox, Git, LaTeX

Research Methods: Multi-agent and constrained reinforcement learning, Lagrangian methods, CTDE, wireless system simulation, multimodal perception

Languages: Mandarin Chinese (native), English (TOEFL iBT 95)